

PUSHPAK JAISWAL

Software Engineer

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OBJECTIVE

Full-Stack Software Developer and Open-Source contributor with a proven record of shipping production systems across microservices, ML pipelines, and CLI tooling, recognized in multiple national level hackathons and competitions.

TECHNICAL COMPETENCIES

- **Languages** : - Java, Python, Vlang (V), JavaScript, TypeScript, C, R, Bash
- **Frameworks** : - Spring Boot, NestJs, FastAPI, React, React Native (Expo), Tailwind CSS, TensorFlow, Scikit-learn, Vweb (Vlang web framework)
- **Tools & DevOps** : - Git, GitHub, Docker, Linux (ubuntu), Vim, Kafka, Maven, Gradle, NPM
- **Cloud & CI/CD** : - GitHub Actions, Docker Hub, GraalVM
- **Testing** : - Junit, Postman, H2, Spring Security
- **Databases** : - SQL, MySQL, PostgreSQL, Redis, MongoDB, CouchDB, ChromaDB

PROFESSIONAL EXPERIENCE

CONTRIBUTOR & MAINTAINER | IBM - AssetOpsBench | Remote | [\[Link\]](#) 2025 – Present

Maintaining high-scale open-source benchmarking tools focused on Facility Management and IoT domain scenarios.

- **Optimized data reliability** for the AssetOpsBench repository (1.3k+ GitHub stars) by performing rigorous data verification, ensuring high-quality benchmarks for FMSR and IoT model training.
- **Scaled technical engagement** by administering a Codabench competition for 1,000+ participants, managing technical evaluations and the distribution of INR 100,000 in prize funding.
- **Enhanced project sustainability** through the design of structured onboarding processes, successfully managing the end-to-end contributor lifecycle for a growing global community.

CO-FOUNDER & CONTRIBUTOR | TechAmigo (Open Source) | Remote | [\[Link\]](#) 2025 – Present

Leading an open-source organization dedicated to building high-impact developer productivity and competitive programming tools.

- **Reduced backend development overhead** by architecting the [create-server-startup](#) NPM package, supporting 4 database integrations and JWT authentication for 50+ weekly active users.
- **Optimized platform security** for the [CPAmigo](#) competitive coding platform, implementing zero-configuration protocols (Helmet, Rate Limiting, CORS) for a user base of 500+ registered members.
- **Managed project lifecycles** for multiple high-impact tools, overseeing strategic growth from initial scaffolding to full-scale deployment using TypeScript and JavaScript.

SOFTWARE & ML INTERN | PWSkills | Remote | [\[Link\]](#) Sep 2024 – Nov 2025

Developed and deployed production-grade full-stack microservices and machine learning models for educational technology.

- **Engineered 7 production-grade systems**, including 3 Spring Boot microservice backends (RESTful APIs, Hibernate, MongoDB) and 4 ML models (TensorFlow, XGBoost) to streamline application workflows.
- **Secured full-stack interfaces** by building responsive React.js and Angular frontends while validating 15+ API endpoints via Spring Security and Postman with H2 databases.
- **Established curriculum benchmarks** after 2 primary projects were selected for use across PW Skills cohorts, directly influencing the learning standards for over 5,000 students.

EDUCATION

B.TECH - CSE | CSMSS CSCOE | 2023-2027

CGPA: 7.5

HSC - SCIENCE | DEOGIRI JUNIOR COLLEGE | 2021-2023

PERCENTAGE: 68.5%

ICSE - 10TH | PODAR INTERNATIONAL SCHOOL | 2011-2021

PERCENTAGE: 89%

PROJECTS

PUSHCODE | [\[Link\]](#)

Technologies: - Spring Boot, Docker, LangChain4j, React, MongoDB

A distributed, AI-enhanced code editor featuring secure, sandboxed execution and real-time diagnostics.

- **Engineered a 4-service microservice architecture** (Security, Execution, AI Agent, Eureka) utilizing Docker-in-Docker (DinD) for sandbox isolation, achieving zero host breaches in production environments.
- **Integrated an agentic Groq pipeline** to auto-detect languages and stream error-correcting code via WebSockets, maintaining <150ms latency for seamless user interaction.
- **Optimized performance using GraalVM** native builds to cut service startup times by ~40%, leveraging Eureka discovery to enable independent, high-availability scaling of individual services.

AGROML-HUB | [\[Link\]](#)

Technologies: - FastAPI, TensorFlow, Spring Boot, React, XGBoost

A polyglot machine learning platform for comparative analysis and edge deployment across agricultural domains.

- **Benchmarked 4 distinct ML models** and implemented TF-Lite edge deployment, which reduced inference overhead by ~60% for users on low-resource devices.
- **Architected a dual-backend system** using FastAPI for high-speed prediction delivery (<200ms) and Spring Boot for robust user authentication and management.
- **Ensured project scalability and security** by managing 95+ pull requests with enforced security and code-quality gates throughout the full project lifecycle.

AWESOME INDUSTRIAL DATASETS | [\[Link\]](#)

Technologies: - Python, LLMs, GitHub Actions, Data Engineering

One of the largest open-source industrial AI collections, featuring automated data engineering pipelines.

- **Indexed and categorized 176+ datasets** across 10 specialized domains including Aerospace and Semiconductors, creating a centralized hub for industrial AI research.
- **Developed an LLM-powered metadata extractor** (llm_automate.py) that boosted tagging throughput by ~40%, significantly decreasing manual review time for new contributions.
- **Automated deployment workflows** using GitHub Actions CI to generate a searchable HTML interface and standardized CSV reports upon every project update.

CERTIFICATIONS

Deep Learning – IIT Ropar (NPTEL) [\[Link\]](#)

Design & Analysis of Algorithms – IIT Kharagpur (NPTEL) [\[Link\]](#)

Advanced Java – GeeksforGeeks [\[Link\]](#)

Software Engineer – HackerRank [\[Link\]](#)

AWARDS

- **1st Place, National Paper Presentation** | K.K. Wagh, Nashik (2026)
- **1st Place, National Project Presentation** | K.K. Wagh, Nashik (2026)
- **4th Place & Yuva Bharat Award** | National Ideathon, GCE Aurangabad (2026)
- **3rd Place, TechnoSpark Hackathon** | MIT, CSN (2025)
- **3rd Place, Supernova Stellar Hackathon** | MGM University (2025)